

1. SEMESTER 1

a. Paper 1: Foundations

- i. Foundation Week
- ii. Python Programming
- iii. NumPy, Pandas & Matplotlib
- iv. Mathematics, Statistics & Probability
- v. Applied Mathematics for Machine Learning

b. Paper 2: Data Engineering

- i. Database & SQL
- ii. Advanced SQL & Data Modeling
- iii. Data Wrangling & Preparation
- iv. EDA & Visualization

c. Paper 3: Machine Learning

- i. Introduction to Machine Learning
- ii. Supervised Learning
- iii. Unsupervised Learning, Feature Engineering & Model Optimization

d. Paper 4: ML Engineering

- i. ML API Design & Development
- ii. Containerization & Cloud for ML

2. SEMESTER 2

a. Paper 5: Deep Learning

- i. Introduction to Deep Learning with TensorFlow/Keras
- ii. Computer Vision & CNN
- iii. Advanced Deep Learning (RNN, LSTM, Attention)
- iv. Time Series Analysis

b. Paper 6: NLP & Generative AI

- i. NLP Fundamentals
- ii. Transformer Architecture
- iii. Generative AI & LLM Fundamentals
- iv. Agentic AI, MCP & AI Engineering

c. Paper 7: MLOps

- i. MLOps
- ii. Production ML Systems & Capstone Project Planning

d. Paper 8: Projects & Placement

- i. Capstone Project Planning
- ii. Capstone Project Development

- iii.

Placement Training 1
- iv.

Placement Training 2

Complete Structure at a Glance

Semester	Paper	Subject	Modules
1	Paper 1	Foundations	5
1	Paper 2	Data Engineering	4
1	Paper 3	Machine Learning	3
1	Paper 4	ML Engineering	2
2	Paper 5	Deep Learning	4
2	Paper 6	NLP & Generative AI	4
2	Paper 7	MLOps	2
2	Paper 8	Projects & Placement	4
		Total	8 papers and 28 modules